BSAN 400 Introduction to Machine Learning

Lecture 4. Logistic Regression and Classifications[1]

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[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

From Continuous to Categorical Outcome

- The response variable, Y, is categorical
- Examples:
  - Banking: default vs. nondefault
  - Medical: disease positive vs. negative
  - Computer vision: sign recognition by self-driving cars
  - Many others...

Classification

- Denote $\mathcal{C}(X)$ as a classifier
- Most DM algorithms estimate the probability that X belongs to each class
- Based on specific decision rules, classification can be produced
- Example: The model prediction tells that the probability of default is 0.2, then

Threshold	<0.1	>0.1
Class	Nondefault	Default

Classification Methods

- K-Nearest Neighbor
- Logistic regression
- Classification tree
- Discriminant analysis
- Support vector machine
- Neural networks
- Deep learning
- ...
- Is clustering a classification model?

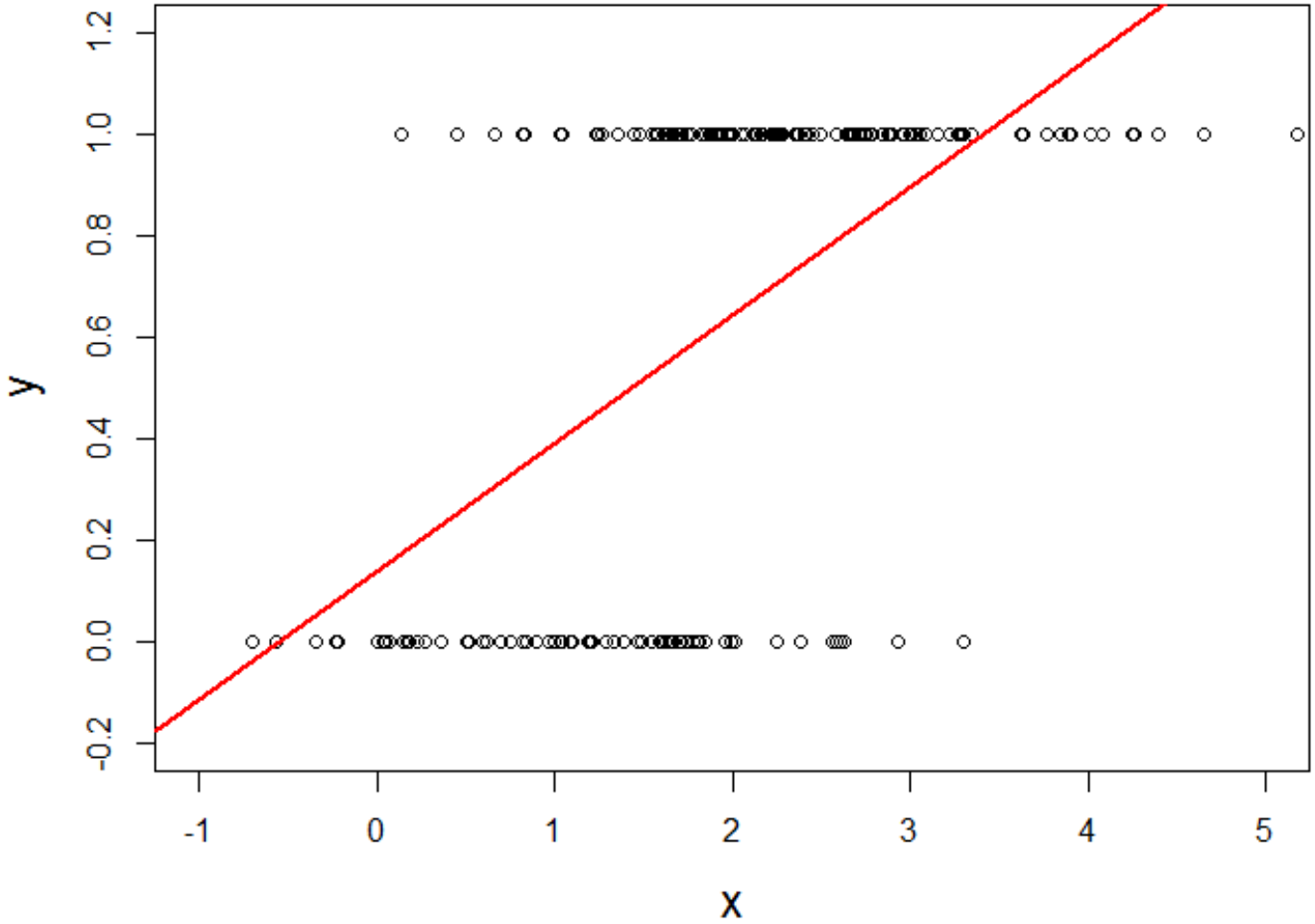
Why Not Linear Regression

- Example: default prediction
  - Default ($Y = 1$) vs. Nondefault ($Y = 0$)
  - X_1: credit card balance level, X_2: income level
- Suppose the estimated linear regression is

$$\hat{Y} = -1.5 + 2X_1 - X_2$$

- What is the predicted value if a person's balance level is 1 and income level is 3?
- How to interpret this value?

An Illustration



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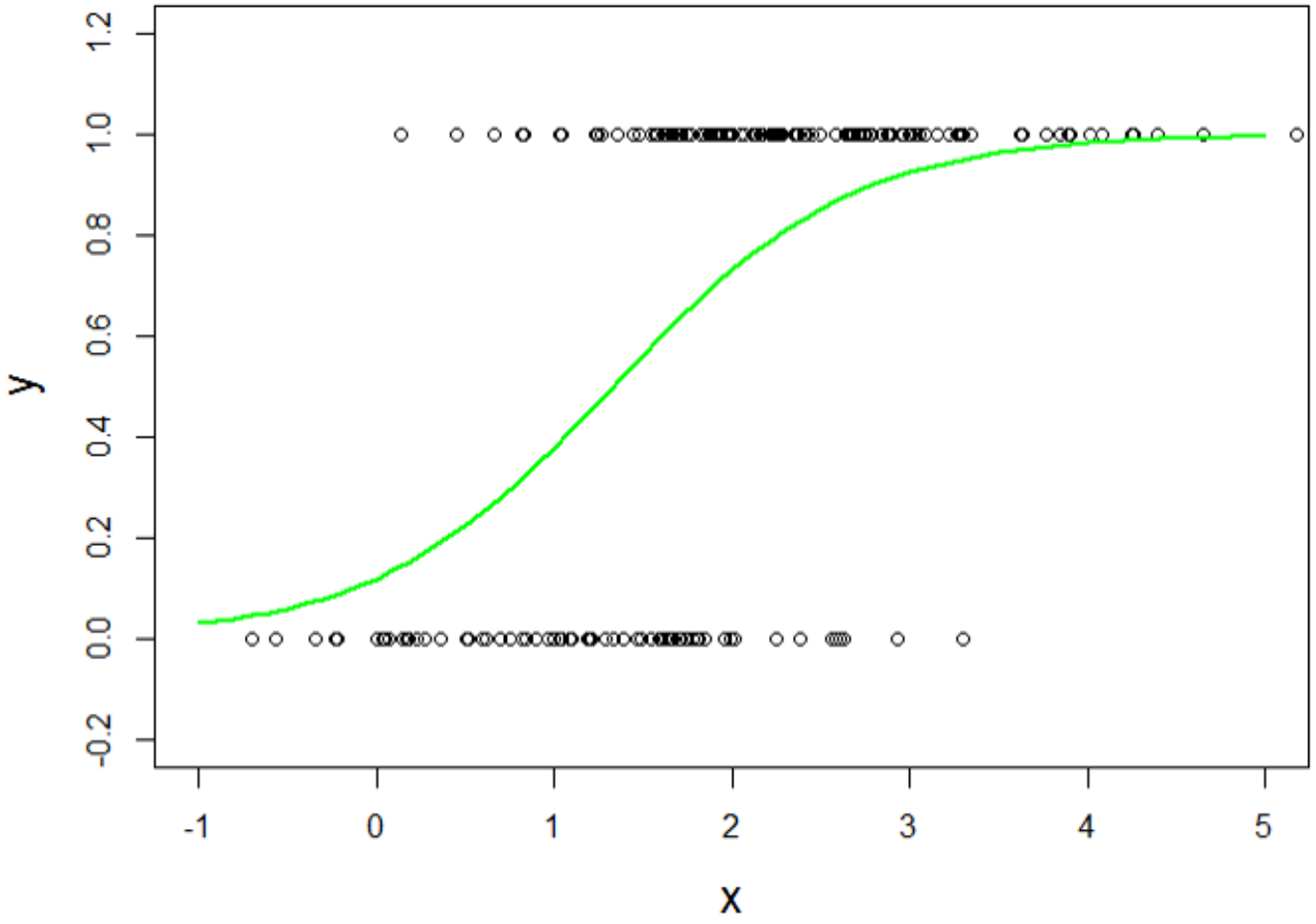


Logistic Regression

- Generalized linear model
- Logistic model for binary response

$$\mathbb{P}(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + \exp(-\beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})}$$

- Sigmoid function
- Function of X
- Outcome is **predicted probability** of event
- More than two classes: *multinomial logistic model*

Loss Function (in machine learning)

- Recall LS

$$L_{OLS}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- For logistic regression, we have a different loss

$$L_{logit}(\boldsymbol{\beta}) = \sum_{i=1}^{n} -2 \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]$$

where $p_i = \frac{1}{1+\exp(-\beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})}$

Maximum Likelihood Estimation (in statistics)

- Maximum likelihood estimation
- $y_i|\mathbf{x}_i \sim \text{Ber}(p_i(\mathbf{x}_i))$
- Likelihood function of $y_i|\mathbf{x}_i$:

$$
\begin{aligned}
L(y_i; \mathbf{x}_i, \boldsymbol{\beta}) &= p_i^{y_i} \times (1 - p_i)^{1-y_i} \\
&= \left(\frac{\exp(\mathbf{x}_i^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta})} \right)^{y_i} \times \left(\frac{1}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta})} \right)^{1-y_i}
\end{aligned}
$$

- By simple algebra, total log-likelihood is (show in exercise)

$$
\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left\{ y_i \mathbf{x}_i^T \boldsymbol{\beta} - \log \left(1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta}) \right) \right\}
$$

How to solve for β?

- No analytical solution like OLS
- Solve numerically (univariate example)
  - Gradient descent

$$\hat{\beta}^{(n+1)} = \hat{\beta}^{(n)} - \alpha * L'(\hat{\beta}^{(n)})$$

    where α is called learning rate.
  - Newton's method (a very good tutorial)

$$\hat{\beta}^{(n+1)} = \hat{\beta}^{(n)} - \frac{L'(\hat{\beta}^{(n)})}{L''(\hat{\beta}^{(n)})}$$

    - We call derivative for univariate: $f'(x)$, $f''(x)$
    - We call gradient for multivariate: $\nabla f(x)$, $\nabla^2 f(x)$

Odds and Interpretation of β

- Logistic model is also called log odds model
- Odds: ratio of probabilities:

$$\text{Odds}(X) = \mathbb{P}(Y = 1|X)/\mathbb{P}(Y = 0|X)$$

- Logit link (logit transformation)

$$\text{logit}(P) = \log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

- By simple algebra, given all X's are fixed except X_j

$$\beta_j = \log\left(\frac{\text{Odds}(X_j + 1)}{\text{Odds}(X_j)}\right)$$

which is *log of odds ratio*.

Multinomial Logit Model

- Response $Y = 1, 2, ..., K$, K classes
- Given predictors $\mathbf{x}_i$

$$\log\left(\frac{\mathbb{P}(y_i = 2)}{\mathbb{P}(Y_i = 1)}\right) = \boldsymbol{\beta}_2^T \mathbf{x}_i$$
$$\log\left(\frac{\mathbb{P}(y_i = 3)}{\mathbb{P}(Y_i = 1)}\right) = \boldsymbol{\beta}_3^T \mathbf{x}_i$$
$$\vdots$$
$$\log\left(\frac{\mathbb{P}(y_i = K)}{\mathbb{P}(Y_i = 1)}\right) = \boldsymbol{\beta}_K^T \mathbf{x}_i$$

- The first class "1" is the reference
- There are $(K-1) \times (p+1)$ coefficients need to be estimated.

Prediction — From Probability to Class

- Direct outcome of model: probability
- Next step: classification
- Need decision rule (cut-off probability – p-cut)
- Not unique

Confusion Matrix

- Classification table based on a specific cut-off probability
- Used for model assessment

	Pred=1	**Pred=0**
True=1	True Positive (TP)	False Negative (FN)
True=0	False Positive (FP)	True Negative (TN)

- FP: type I error; FN: type II error
- Different p-cut results in different confusion matrix
- Try to understand this table instead of memorizing!

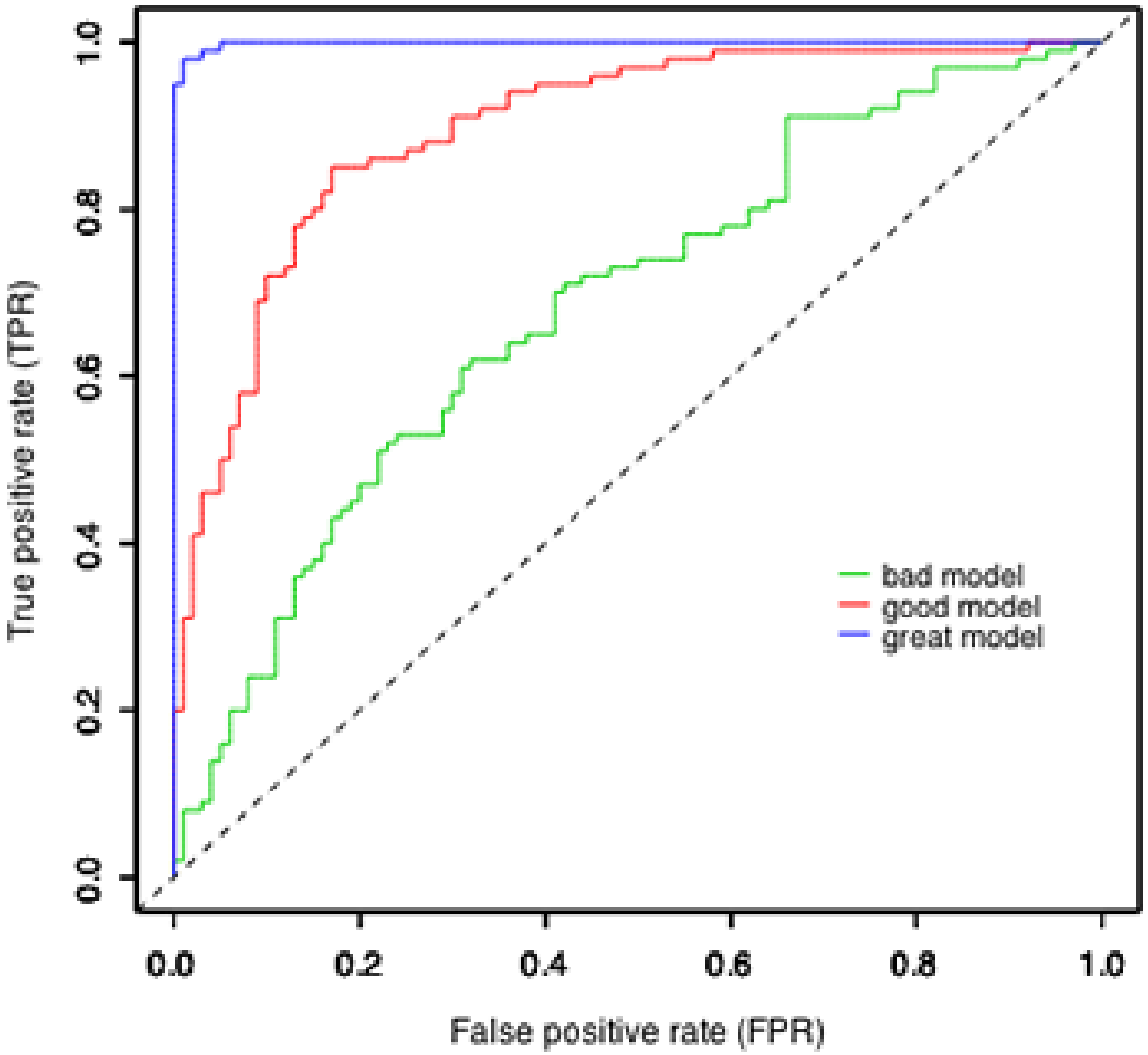
Some Useful Measures

- Misclassification rate (MR) $= \frac{\text{FP+FN}}{\text{Total}}$
- True positive rate (TPR) $= \frac{\text{TP}}{\text{TP+FN}}$: Sensitivity or Recall
- True negative rate (TNR) $= \frac{\text{TN}}{\text{FP+TN}}$: Specificity
- False positive rate (FPR) $= \frac{\text{FP}}{\text{FP+TN}}$: 1 – Specificity
- True negative rate (FNR) $= \frac{\text{FN}}{\text{TP+FN}}$: 1 – Sensitivity
- Positive predictive rate (PPR) $= \frac{\text{TP}}{\text{TP+FP}}$: Precision
- False discovery rate (FDR) = 1 – Precision

ROC Curve

- Receiver Operating Characteristic
- Plot of FPR (X) against TPR (Y) at various p-cut values
- Overall model assessment (not for a particular decision rule)
- Unique for a given model
- Area under the curve (AUC): a measure of goodness-of-fit

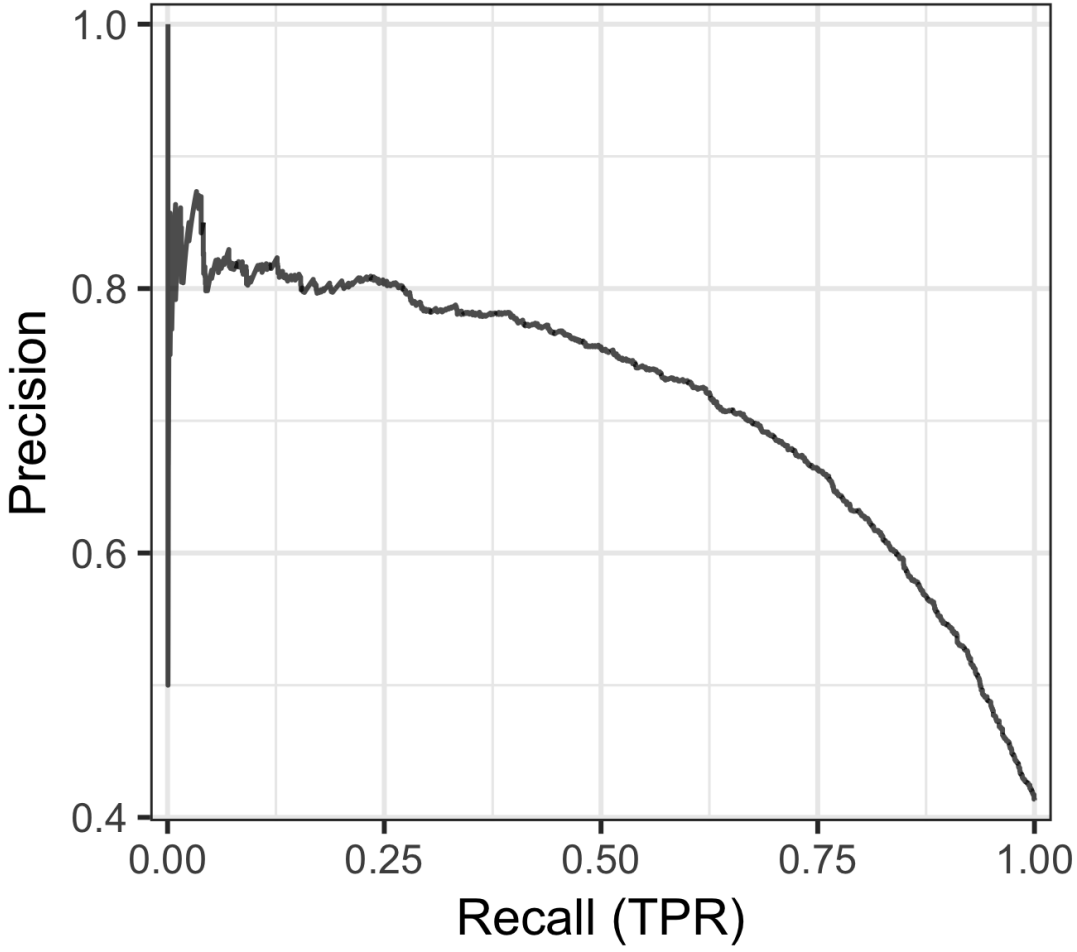
ROC Curve


True positive rate (TPR)
False positive rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
bad model
good model
great model

Precision and Recall

- More accurate measure for imbalanced data
- Widely used in document retrieval
- Precision $= \frac{\text{TP}}{\text{TP+FP}}$:
  - fraction of retrieved instances that are relevant
- Recall $= \frac{\text{TP}}{\text{TP+FN}}$:
  - fraction of relevant instances that are retrieved
- Neither incorporates TN ($Y = 1$ is of more interest)
- F-score: $F = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
- More details: see this highly cited paper

Precision-Recall Curve



Asymmetric Cost

- Example: compare following two confusion matrices based on two p-cut values

	Pred=1	**Pred=0**
True=1	10	40
True=0	10	440

	Pred=1	**Pred=0**
True=1	40	10
True=0	130	320

- Which one is better? In terms of what?
- What if this is about loan application
  - $Y = 1$: default customer
  - Default will cost much more than reject a loan application

Choice of Decision Threshold (p-cut)

- Do NOT simply use 0.5!
- In general, we use grid search method to optimize a measure of classification accuracy/loss
  - Cost function (symmetric or asymmetric)
  - F-score based on precision and recall
- Grid search with cross-validation